Hints: Linear algebra – 3. Eigenvalues, diagonalisability and the Jordan form

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. A real matrix with only real eigenvalues can be triangularised by an orthogonal matrix (real Schur form): $A = QTQ^T$ with T upper triangular. Rewrite the equation in terms of T.

Hint 2. $T + T^k = T^T$: the left side is upper triangular and the right side is lower triangular, so T is diagonal. Then $T^k = O$ for a diagonal T. Answer: only $A = O$.

Problem 2

Hint 1. $A^m = 2021B - B^2$ is symmetric, and its eigenvalues are $2021\beta - \beta^2 \leqslant 2021^2/4$. Use this for even exponents.

Hint 2. $A^{2m} = (A^m)^2$ is the square of a symmetric matrix, so its eigenvalues lie in $[0, 2021^2/4]$. Then $|\det A|^{2m} = \det A^{2m} \leqslant (2021^2/4)^n$ for every m; let $m \to \infty$.

Problem 3

Hint 1. Compute A^4 using $A^2 = A^T$ twice.

Hint 2. $A^4 = (A^T)^2 = (A^2)^T = A$, so every eigenvalue satisfies $\lambda^4 = \lambda$: $\lambda = 0$ or $\lambda^3 = 1$. All four values occur: O, E, and the 2×2 rotation by $2\pi/3$ (for which $R^T = R^{-1} = R^2$). Answer: $\lambda \in \{0, 1, e^{\pm 2\pi i/3}\}$.

Problem 4

Hint 1. Write the rotation as a matrix product: with J the anti-diagonal permutation matrix (ones on the anti-diagonal), $A^R = JA^T$. So the condition is $A = JA^T$, i.e. $A^T = JA$, and transposing, $A^T = AJ$.

Hint 2. Hence $JA = AJ$, and $A^TA = JA^2 = AA^T$: A is normal. For a normal real matrix, $Av = \lambda v$ implies $A^Tv = \bar{\lambda}v$.

Hint 3. Then $\bar{\lambda}v = A^Tv = JAv = \lambda Jv$. If $\lambda \neq 0$, v is an eigenvector of J with eigenvalue $\bar{\lambda}/\lambda$, and $J^2 = E$ forces $(\bar{\lambda}/\lambda)^2 = 1$, i.e. $\bar{\lambda} = \pm\lambda$.

Problem 5

Hint 1. Factor $3x^3 - x^2 - x - 1 = (x - 1)(3x^2 + 2x + 1)$.

Hint 2. The roots are 1 and two complex roots of modulus $1/\sqrt{3}$, all simple. So A is diagonalisable over $\mathbb{C}$ with eigenvalues among these. Hence A^k converges to the projection onto the 1-eigenspace along the others, which is idempotent (and real, as a limit of real matrices).

Problem 6

Hint 1. A rank-one matrix is $C = uv^T$, and $C^2 = (v^Tu)C = (\operatorname{tr} C)\, C$. What is $\operatorname{tr}(AB - BA)$?

Problem 7

Hint 1.

(a) The polynomial $x^3 - x - 1$ has a real root r. Try a scalar matrix.

Hint 2.

(b) The eigenvalues of A are roots of $x^3 - x - 1$. This polynomial has exactly one real root, and it is positive (study f' and $f(0)$). Since A is real, its non-real eigenvalues come in conjugate pairs with equal multiplicities, so $\det A = r^a|z|^{2b} > 0$.

Problem 8

Hint 1.

(a) The minimal polynomial divides $x^2 - 1 = (x-1)(x+1)$, which has simple roots (over $\mathbb{R}$ or $\mathbb{C}$, characteristic $\neq 2$). Explicitly, $v = \frac{v+Av}{2} + \frac{v-Av}{2}$.

Hint 2.

(b) Commuting diagonalisable operators are simultaneously diagonalisable: each preserves the eigenspaces of the others. In a common eigenbasis every involution is $\mathrm{diag}(\pm 1, \ldots, \pm 1)$. Answer: 2^n.

Problem 9

Hint 1. Write $Y = XN$, where N is the shift with $Ne_j = e_{j+1}$ for $j < n$ and $Ne_n = 0$.

Hint 2. Then $B = X^{-1}Y = N$ and $A = YX^{-1} = XNX^{-1}$. Both are similar to the nilpotent Jordan block N, which has rank $n - 1$.

Problem 10

Hint 1. The condition says $(1, \ldots, 1)A = \lambda(1, \ldots, 1)$, so λ is an eigenvalue of A^T. Now use $\chi_{A^T} = \chi_A$.

Problem 11

Hint 1. $\operatorname{rank} A \neq \operatorname{rank} A^2$ means $\ker A \subsetneq \ker A^2$: there is x with $A^2x = 0$ but $y = Ax \neq 0$. Look for B of rank one, $B = yz^T$.

Hint 2. Take $z \neq 0$ with $z^TA = 0$ (it exists since A is singular). Then $AB = (Ay)z^T = O$, $BA = y(z^TA) = O$ and $B^2 = (z^Ty)B$ with $z^Ty = z^TAx = 0$.

Problem 12

Hint 1. Since A is not scalar, some v is not an eigenvector, so v, Av are independent. In a basis starting with v, Av the first column is $(0, 1, 0, \ldots, 0)^T$; in particular the $(1,1)$ entry is 0. Prove by induction the stronger claim that the diagonal can be made $(0, \ldots, 0, \operatorname{tr} A)$.

Hint 2. Let A' be the lower-right $(n-1) \times (n-1)$ block. Conjugating by $\mathrm{diag}(1, P)$ keeps the $(1,1)$ entry 0 and replaces A' by $P^{-1}A'P$. So if A' is not scalar, you are done by induction.

Hint 3. If $A' = cE$ (and $n \geqslant 3$), change the basis $v, Av, w_3, \ldots, w_n$ to $v, Av, w_3 + v, w_4, \ldots$. The first column is unchanged, and the new block becomes cE plus a non-zero off-diagonal entry, so it is no longer scalar. The base case $n = 2$ is the companion matrix in the basis v, Av.

Problem 13

Hint 1. C is nilpotent if and only if $\chi_C(x) = x^n$. Each coefficient of $\det(xE - A - tB)$ is a polynomial in t of degree at most n.

Hint 2. These polynomials vanish at $n + 1$ points (except the leading x^n), so they vanish identically: $A + tB$ is nilpotent for all t. Hence $(A + tB)^n = O$ for all t. Look at the coefficients of t^0 and t^n.

Problem 14

Hint 1. The coefficients of χ_A are, up to sign, the sums of principal minors, so $\chi_A = x^n$ and $A^n = O$ by Cayley–Hamilton. For the permutation, consider the directed graph with an edge $j \to k$ whenever $a_{jk} \neq 0$.

Hint 2. It suffices to show this graph has no directed cycles: then a topological order of the vertices gives σ. Diagonal entries are 1×1 minors, so there are no loops.

Hint 3. Take a shortest directed cycle $j_1 \to \cdots \to j_m \to j_1$. A chord would create a shorter cycle, so the induced subgraph on these vertices is exactly this cycle. In the expansion of the principal minor on $\{j_1, \ldots, j_m\}$ only one permutation contributes. So the minor is $\pm \prod a_{j_i j_{i+1}} \neq 0$, a contradiction.

Problem 15

Note. as stated, (a) is false. For $X = \operatorname{diag}(1, 0)$, any Y with $Y^2 = X^2$ commutes with $X^2 = \operatorname{diag}(1, 0)$, hence is diagonal, hence $Y = \pm X$. (Part (b) as stated is true: for $A = E + E_{12} + E_{23}$, A^2 is a single Jordan block with eigenvalue 1, and such a matrix has only the two square roots $\pm A$.) The original SEEMOUS 2016 problem reads: (a) for every $X \in M_2(\mathbb{C})$ there is Y with $Y^3 = X^2$; (b) there is $A \in M_3(\mathbb{C})$ with $Z^3 \neq A^2$ for all Z. The hints are for this version.

Hint 1.

(a) Reduce to the Jordan form of X. If X is diagonalisable, take cube roots of the eigenvalues of X^2. If $X = \lambda E + N$ with $N \neq O$, $N^2 = O$, then $X^2 = \lambda^2 E + 2\lambda N$.

Hint 2. For $\lambda = 0$ take $Y = O$. For $\lambda \neq 0$ look for $Y = cE + dN$: $Y^3 = c^3 E + 3c^2 dN$, so solve $c^3 = \lambda^2$, $3c^2 d = 2\lambda$.

Hint 3.

(b) Take A the nilpotent 3×3 Jordan block, so $A^2 \neq O$ but $A^3 = O$. If $Z^3 = A^2$, then $Z^6 = O$, so Z is nilpotent, and in size 3 this forces $Z^3 = O$.

Problem 16

Hint 1. Take n of the eigenvectors as a basis, $Av_i = \lambda_i v_i$, and write the last one as $v_{n+1} = \sum c_i v_i$. Why are all $c_i \neq 0$?

Hint 2. Apply A: $\sum c_i \lambda_i v_i = \mu \sum c_i v_i$, so every $\lambda_i = \mu$. Answer: yes, $A = \mu E$.